📘 Django Interview Notes (Day 2)

🧠 1. What is ORM?

ORM (Object Relational Mapping) is a way to interact with the database using Python code instead of writing raw SQL queries.

👉 Example:

Task.objects.all()

Instead of:

SELECT * FROM tasks;

❓ 2. Why use ORM instead of raw SQL?

- Cleaner and more readable code
- Database independent (works with PostgreSQL, MySQL, SQLite)
- Prevents SQL injection attacks
- Faster development

👉 In interviews:
ORM improves productivity and security.

🔄 3. What are Migrations?

Migrations are Django's way of applying changes in models to the database.

👉 Example:

- Add field → migration updates DB
- Create table → migration runs SQL

⚙️ 4. Difference: makemigrations vs migrate

Command	Purpose
makemigrations	Creates migration files
migrate	Applies changes to database

👉 Simple:

- makemigrations = prepare
- migrate = execute

📦 5. What is a QuerySet?

A QuerySet is a collection of database queries.

👉 Example:

Task.objects.all()

👉 It is lazy (executes only when needed)

🧩 6. What is a Model in Django?

A model is a Python class that represents a database table.

👉 Each field = column
👉 Each object = row

🖨️ 7. What is str()?

It defines how an object is displayed (especially in admin panel).

👉 Example:

```
def __str__(self):
    return self.title
```

🔑 8. What is Primary Key?

A primary key uniquely identifies each record in a table.

👉 Django automatically creates:

id = AutoField(primary_key=True)

🆔 9. What is default id field?

Django automatically adds an id field as primary key unless specified.

⚖️ 10. null=True vs blank=True

Option	Meaning
null=True	Database level (can store NULL)
blank=True	Form level (field can be empty)

⏱️ 11. auto_now vs auto_now_add

Field	Use
auto_now	Updates every time object saves
auto_now_add	Set only once (creation time)

🧠 12. What is Meta class?

Used to define extra options for model.

👉 Example:

```
class Meta:
    ordering = ['-created_at']
```

⚡ 13. What is db_index?

It creates an index on a field to make queries faster.

👉 Used for frequently searched fields.

🔒 14. What is unique constraint?

Ensures that a field value is unique.

👉 Example:

```
email = models.EmailField(unique=True)
```

🛡️ 15. How Django prevents SQL Injection?

- Uses ORM (no direct SQL queries)
- Escapes user inputs automatically
- Uses parameterized queries internally

👉 This makes Django secure by default.

🧪 Common ORM Operations

Create

```
Task.objects.create(title="Learn Django")
```

Read

Task.objects.all()

Update

task.save()

Delete

task.delete()

📌 **Key Takeaways**

- ORM = Python → Database bridge
- Migrations sync models with DB
- QuerySets are lazy
- Admin panel uses models directly
- Django auto handles security (SQL injection)